

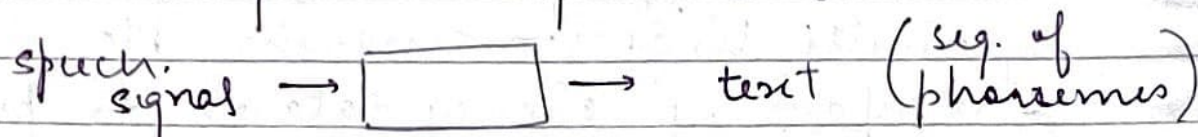
Speech Tech - Week-6

1/1

L1

Review of Speech Feature Extraction

- Speech recogniⁿ → speech to text



very similar to handwriting recogniⁿ.

↳ there is a lot of difference in cursive writing & OCR-editable text.

- seq. of phonemes → lexicon / seq. of words

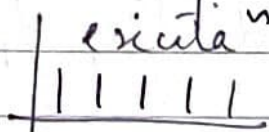
basic units

of sound / spoken language

2 basic categories

Vocal Unvoiced

they have their own resonant frequency



vocal tract

$V(\omega)$

formants

$\Rightarrow S(\omega)$

$= E(\omega) \times$

ω

diff. phonemes have diff. frequency characteristics.

① Mel $\Rightarrow$ mel freq. (used by ear)

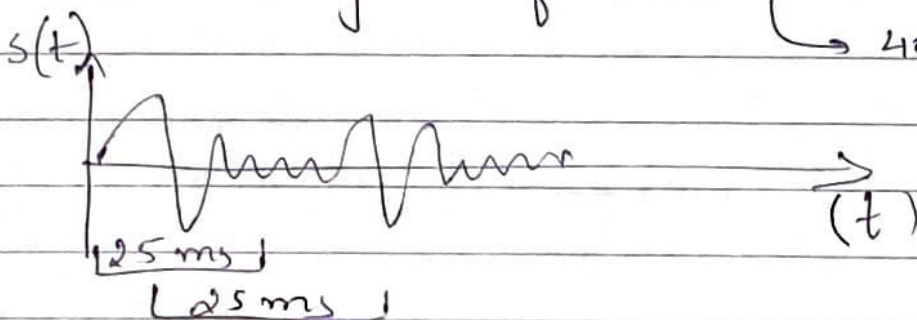
use triangular filters

80

WB

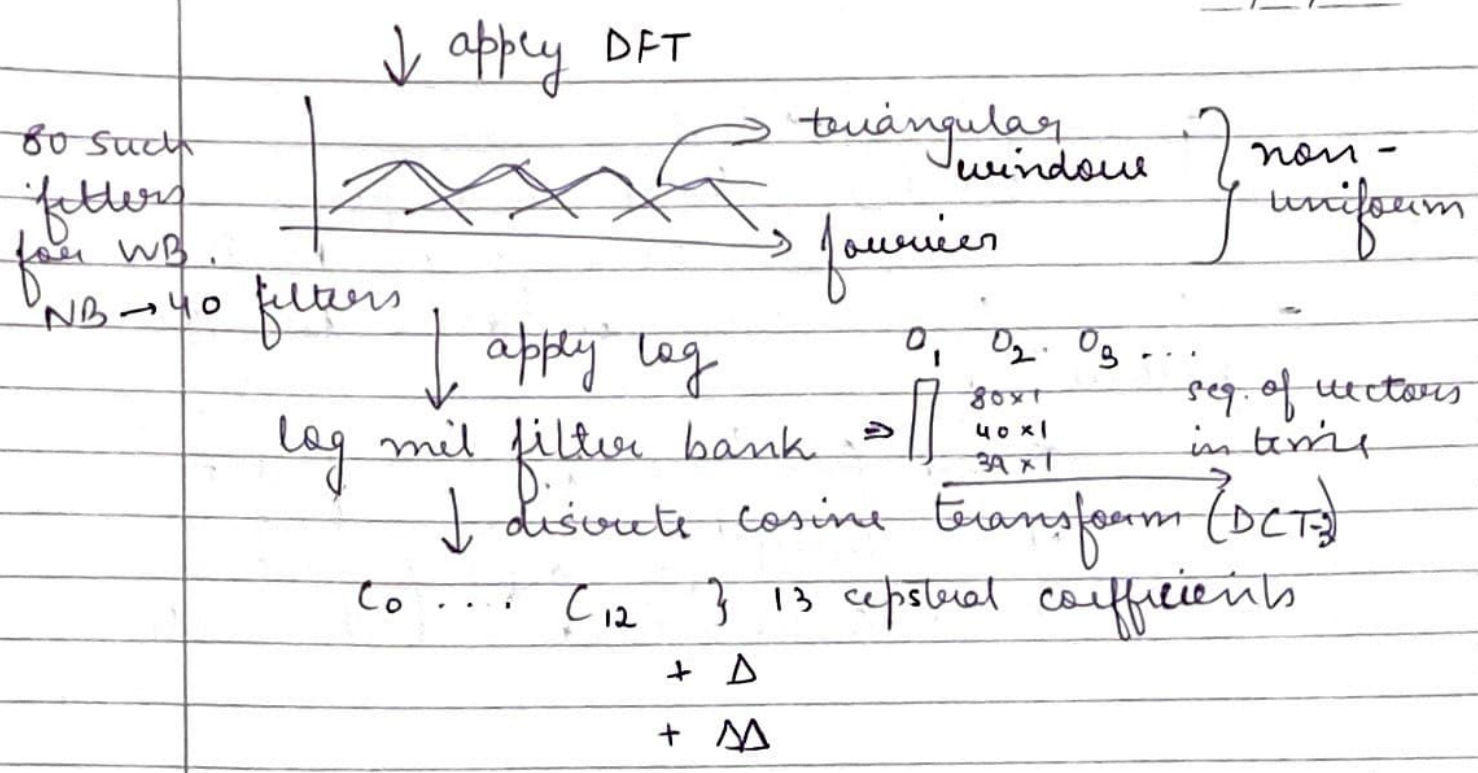
40

NB



window size $\rightarrow$ 25 ms

shift $\rightarrow$ 10 ms



L2 Review of a Pattern Classification Problem

- pattern classification btw. male & female features → ht. & wt ⇒ bivariate measurements of features
- from these build a male & female model

Indian & Non-Indians & Pygmy

$\begin{bmatrix} ht \\ wt \end{bmatrix} \Rightarrow$ children, w, m

bivariate

$$W_C^T N_C^T (x; \mu_C^T, \Sigma_C^T) +$$

$$W_W^T N_W^T +$$

$$W_M^T N_M^T$$

GMM

Similarly GMM for pygmies & non-pygmy people

queen $\begin{bmatrix} \text{ht.} \\ \text{wt.} \end{bmatrix} \rightarrow \text{test}$
 $\hookrightarrow$ is this person
 belongs I, N, P

- similar thing, will be done for speech recogniⁿ as well.

$I \rightarrow \text{GMM}^I(t, \lambda_i)$
 $N \rightarrow \text{GMM}^N(t, \lambda_N)$
 $P \rightarrow \text{GMM}^P(t, \lambda_P)$

} find out where it is more probable.

L6.3 Limited Vocab. Speech Recogniⁿ

- in my lang. there are 2 sounds $\rightarrow /aa/$ & $/iy/$

queen sound file

$\xrightarrow{\text{feature}}$ $\begin{cases} /aa/ \\ /iy/ \end{cases}$ binary classification

extract the feature vectors

MFCC $0, a_2 \dots (31 \times 1)$
 using MFCC features, we will build a model.

$\Rightarrow$ data with labels

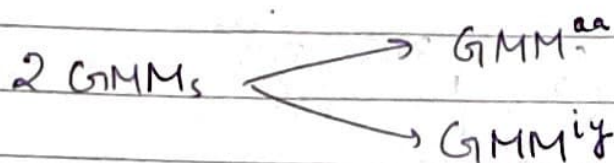
$\rightarrow$ 1000 people say $/aa/$
 $s_1(t) \dots s_{1000}(t) \rightarrow \text{label } /aa/$
 $\downarrow$
 MFCC $\{ \square \dots \square \} \Rightarrow$ features for $/aa/$

$\rightarrow$ 1000 people say $/iy/$

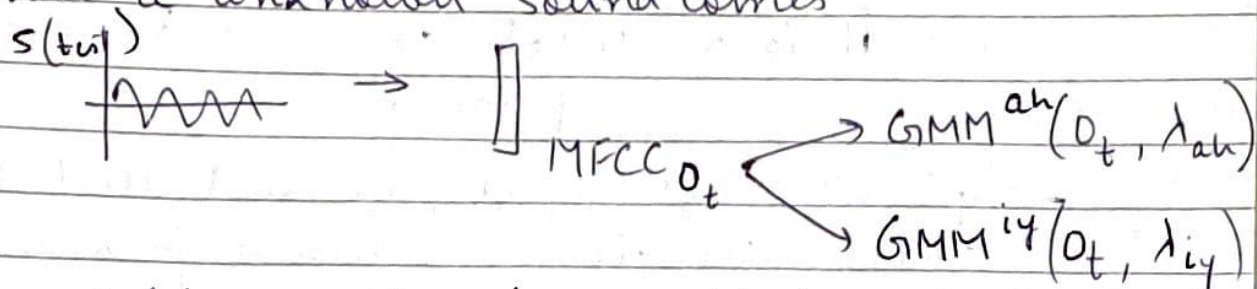
get MFCC vectors with labelled data

- len of vocal tract $C < W < M$

build a GMM $\rightarrow$ multivariate (39×1)



- when a unknown sound comes



whichever is higher, it belongs to that class.

- but speech it is a problem

and $\rightarrow$ $|a| + |m| + |d| \rightarrow$ seq. of sounds
problem

isolate problem $\leftarrow$
be sympathetic towards computers

L6.4 Review of Ltd. Vocab. Speech Recognition

- example recap of I vs N vs P $\rightarrow$ GMM with bivariate distribution ($^{nt}_{wt.}$) for C, W, M .
- then recall of binary classificaⁿ with 3T vs $\Sigma \rightarrow$ multi-variate distribution ($^{80 \times 1}_{40 \times 1}$)

- varying dimension of vocal tract
 - environment noise
 - different transducers

$$\sim \sum_{i=1}^{16} N(x; \mu_i, \Sigma_i)$$

it is modelled as a mixture of gaussians

- given a test signal, we send that to both GMMs with their relevant parameters.
 - in which GMM, I get higher probability.
 - I'll assign my test speech signal to that sound tract.

LS Intro. to HMM GMM Model for Speech Recogⁿ - 1

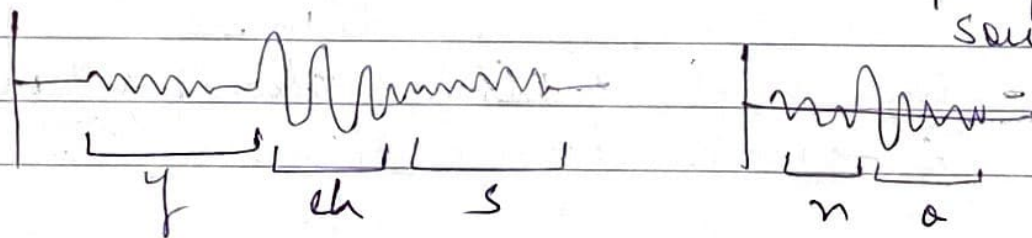
- my new lang. has 2 words → YES & NO

$\left. \begin{matrix} z_1(t) \\ \vdots \\ z_{100}(t) \end{matrix} \right\} \begin{matrix} \text{yes} \\ \downarrow \\ \text{3 sounds} \end{matrix}$

/y/ /eh/ /s/

$\begin{matrix} \text{no} \\ \downarrow \\ \text{2 sounds} \end{matrix}$
 /n/ /a/

we have to deal with HMM → seq. of sounds

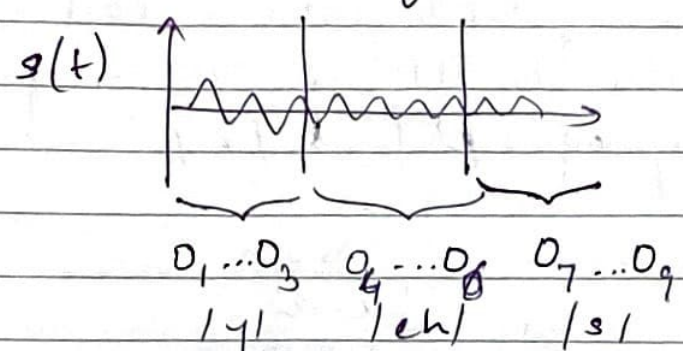


25ms

each 25 ms window
creates an MFCC vector $O_1, O_2, O_3, \dots$
log mel FB vector

sequence of vectors

- yes has 3 sounds $\rightarrow /y/ /eh/ /s/$
 if we knew the boundary of phonemes for each waveform



easily build GMM for each sound.

But, in practice, we generally don't know the boundaries or segments.

- Analogous - to hidden informaⁿ in gaussian mixture models $\Rightarrow$ didn't have labels
 so, we randomly initialize C & P $\Rightarrow$ 2 gaussians and assign each data point to one of the 2 categories. $\Rightarrow$ labelled data
- similarly, we don't know the boundaries of /y/, etc. Recall the ideas used in markov chains.

we still don't know the labels

$s_1(t)$ $\rightarrow$
 $\vdots$ $O_1 \dots O_{94}$
 $s_{1000}(t)$
 $O_1^{1000} \dots O_{76}^{1000}$

$\rightarrow$ uniformly segmented

yes $\rightarrow$ 3 sounds $\rightarrow$ 3 equal intervals

for no. of sounds available.

$$s_1(t) \xrightarrow{\frac{2\pi}{3}} \underbrace{0_1' \dots 0_{31}'}_{\text{'y'}} \quad \underbrace{0_{32}' \dots 0_{63}'}_{\text{'ch'}} \quad \underbrace{0_{64}' \dots 0_{94}'}_{\text{'s'}}$$

$s_{1000}(t) \rightarrow$ do this for all of them.

- using all log MFCC features belonging to $|y| \rightarrow$ we build the GMM for y .
Similarly of $|ch|$ & $|s|$

$s_i(t)$ ~~vector~~ $\rightarrow$ assign each vector to $|y|$ $|ch|$ $|s|$ by comparing their GMMs

like $GMM^y(o_1)$ vs $GMM^{ch}(o_1)$ vs $GMM^s(o_1)$
you will get the new segments, do this till $s_{1000}(t)$ for all of them, you will get updated segments.
 $\rightarrow$ repeat this process & keep updating them.

16

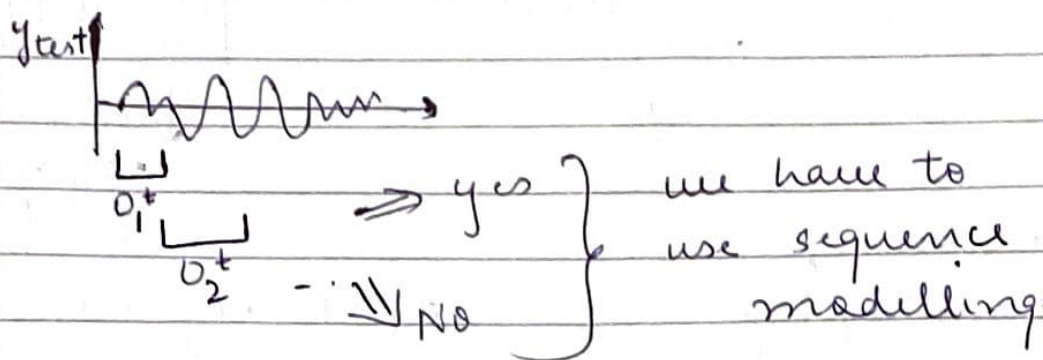
Part - 2

I have,

GMM ^{1y}
GMM ^{1en}
GMM ^{1s}

GMM ¹ⁿ
GMM ^{1o}

- given a test occurrence, classify whether 'yes' or 'no'.



- Recall: HMM, the hidden state was weather condition (for these we have transition probability), then we have observable states (with its observational probability).
- Q $\rightarrow$ which underlying seq. of states (weather condi^{ns}) best explains the observance of day, dayish, day.

Then, we calculated

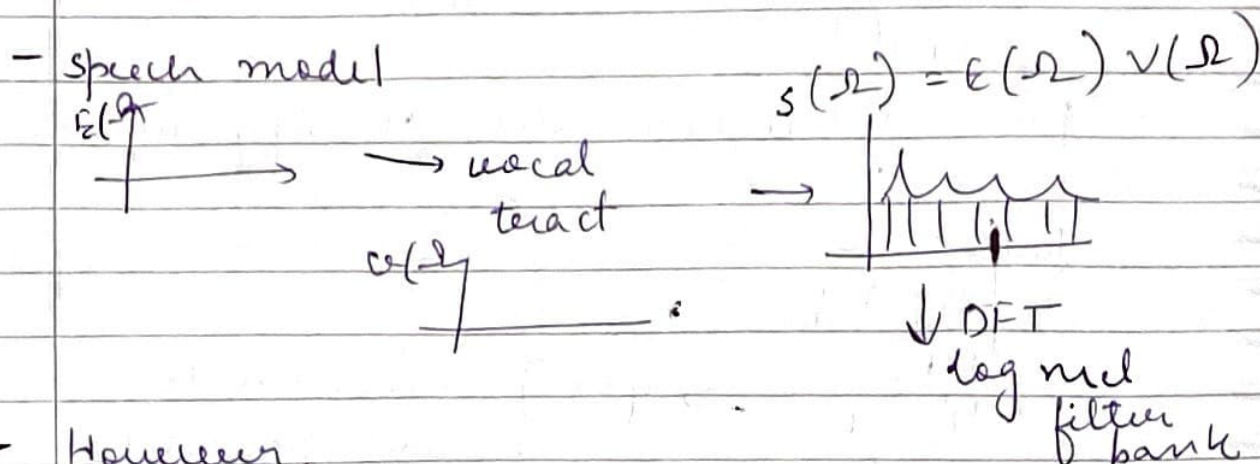
$$\left\{ \begin{array}{l} P(\text{Day} | \text{Sun}) \times P(\text{Sun} | \text{Cloudy}) + \\ P(\text{Dayish} | \text{Sun}) \times P(\text{Sun} | \text{Sun}) + \\ P(\text{day} | \text{cloudy}) \times P(\text{Cloudy} | \text{Sun}) \end{array} \right.$$

this is just 1 path, there will be 27 such paths.

obs x trans

$\therefore$ use viterbi to solve the problem efficiently.

L6.7 HMM Applied to Speech



- However, underlying seq. of sounds are hidden from me.

80x1 log MFB

Observed part

Phonemes $\rightarrow$ are the hidden states

MFBs $\rightarrow$ are the observable states

- /ah/, /ey/, /uh/, /k/, /h/ $\Rightarrow$ these are the phonemes

$$P(o_i | \text{"ah"}) \quad P(\text{"ah"} | \text{"ah"})$$

$\downarrow$
GMM^{ah}

observed probability transition probability $\rightarrow$ the phonemes follows the same.

- In practice, we use transcribed labelled data.

$s_1(t)$  $\xrightarrow{\text{label}}$ How are you?
 $s_2(t)$ - - -

- here, we introduce the concept of lexicon
 $I \rightarrow 'iy'$, $how \rightarrow /h/ /ow/$

(- - - - - so on

↳ this is built once, using the professional

lexicon $\rightarrow$ mapping of words to phonemes
(automatic method like G2P)

$s_1(t) \rightarrow$ Lexicon $\rightarrow$ get corresponding
Seq. of phonemes.

- in the weather example, we were finding the best path of weather possibility using Viterbi.

- $/h/ /ow/ \rightarrow$ you can't go back to $/h/$

$h \rightarrow h$

$ow \rightarrow ow$

ow

sth. new but can't go back

$\square$

o_1

$\square$

o_2

$\square$

o_3

$\square$

o_4

- then, we continue the iterative process.

- GMM models the states

HMM takes care of the Transiⁿ probab (1 phoneme following other phoneme) & also the obs. probab.

Phonemes are the states.

- Forward $\rightarrow$ CTC (connecⁿ temporal classificaⁿ)

↳ DL